

Abhin Binu Kumar

Kerala, India · abhin.b11@gmail.com · linkedin.com/in/abhin-b-a1700121a

Quantitative Research · Financial Modelling · Sector Analysis

PROFESSIONAL SUMMARY

Mathematics and Computing graduate of the Indian Institute of Technology (IIT) Goa (CGPA 8.24/10, academic rank 5) with experience in quantitative financial analysis, multi-year financial modelling, and large-scale data analysis. Prepares feasibility and investment analysis across hospitality, agriculture, healthcare, automotive, and renewable energy, using Python and Excel for scenario analysis, break-even analysis, and debt service coverage ratio (DSCR) sensitivity. Additional experience includes C++ optimisation for operational decision systems and statistical analysis of telecom datasets of more than 6 million customer records. Academic work covers derivatives pricing, Monte Carlo simulation, time-series models, and privacy-preserving synthetic data. Currently advancing expertise in mathematical finance, stochastic modelling, and derivatives pricing through graduate study.

CORE SKILLS

Financial analysis: Financial modelling (modeling), cash-flow analysis, scenario analysis, break-even analysis, DSCR, feasibility analysis, profitability analysis, capital expenditure, investment decision support

Markets and pricing: CAPM, systematic risk, Black–Scholes, option Greeks, geometric Brownian motion, Monte Carlo simulation, hedging sensitivity, volatility, risk management, backtesting

Data and statistics: Statistical modelling, time-series analysis, exploratory data analysis, feature engineering, anomaly detection, churn analysis, forecasting, datasets of 6 million+ records

Optimisation: Branch and Bound, mixed-integer linear programming (MILP), constrained optimisation (optimization), numerical methods, finite element methods, matrix methods

Computing: Python, C++, Microsoft Excel, FreeFem

Machine learning: LSTM sequence models, generative models, differential privacy, PATE, synthetic data

Sectors: Hospitality, agriculture, healthcare, automotive, renewable energy, telecom, logistics and fleet operations

PROFESSIONAL EXPERIENCE

Quantitative Research Associate | Sep 2025 – Present

Neel Corporation, India

- Deliver quantitative financial analysis for live projects in hospitality, agriculture, healthcare, automotive, and renewable energy, and use it in feasibility studies and investment decisions.
- Build multi-year financial models of revenue, operating costs, capital expenditure, profitability, and cash flow, with assumptions written so each case can be traced and rerun.
- Run scenario analysis in Python and Excel on financing assumptions, including break-even points and debt service coverage ratio (DSCR) sensitivity.
- Compare operating and financing cases on profitability and cash flow, and separate the cases that support an investment from the cases that do not.
- Check revenue, cost, and cash-flow lines against each other inside each model before the figures are used in a feasibility discussion.

Research Engineer – Optimisation and Decision Modelling | Sep 2024 – Sep 2025

Addverb Technologies, India

- Worked on large-scale optimisation and constrained decision-making for fleet management systems.
- Formulated fleet assignment as a constrained optimisation problem for allocation decisions across the fleet.
- Designed an optimal assignment algorithm using Branch and Bound.
- Built the cost function from distance, congestion, and fairness, so all three entered the same objective.
- Implemented the optimisation pipeline in C++, from cost evaluation through the search to the chosen assignment.
- Developed pruning strategies that discarded branches unable to improve the best feasible assignment.
- Replaced the previous MILP-based solver with this pipeline and improved system throughput by 18%.
- Analysed the optimality–efficiency trade-off between the Branch and Bound solver and the MILP solver.

Machine Learning Research Intern | May 2023 – Dec 2023

India Urban Data Exchange (IUDX)

- Designed a PATE-GAN pipeline that combines Private Aggregation of Teacher Ensembles with a generative model to produce synthetic datasets under a formal differential-privacy guarantee.
- Measured the privacy–utility trade-off with statistical and distributional metrics on sensitive datasets, including finance-related data.
- Worked with researchers on responsible use of the synthetic data, so later analysis could proceed without exposing the underlying records.
- Presented the results at the Symposium on Data for Public Good, Indian Institute of Science (IISc) Bangalore.

Linear Algebra Tutor | Jan 2023 – Feb 2023

Unacademy

- Taught vector spaces, eigenvalues, eigenvectors, and matrix factorisation.
- Worked through eigenstructure and matrix-decomposition problems so students could apply the methods, not only recall the definitions.

Data Analyst Intern | Jul 2022 – Aug 2022

6D Technologies, India

- Analysed churn across more than 6 million customer records using statistical modelling, exploratory data analysis, and time-series methods.
- Built Python pipelines for feature analysis and anomaly detection on high-volume, noisy telecom data.
- Separated unusual customer behaviour from noise in that telecom data so churn patterns could be reviewed across the full customer base.

SELECTED PROJECTS

Finite Element Methods (FEM) for Solving Partial Differential Equations | Dec 2023 – May 2024

Academic Project, IIT Goa

- Implemented Finite Element Method formulations in FreeFem to solve partial differential equations, modelling systems such as heat transfer and electrostatics.
- Assembled and decomposed large linear systems, then ran stability, convergence, and error analysis on the numerical schemes.

Algorithmic Trading System with Machine Learning | Dec 2023 – Apr 2024

IIT Goa

- Designed an end-to-end algorithmic trading system covering data ingestion, feature engineering, signal generation, execution, and risk management.
- Implemented LSTM-based sequence models for financial time series and integrated them with volatility-adjusted position sizing, stop-loss rules, and drawdown constraints.
- Built a modular Python backtesting pipeline to evaluate strategies across different market conditions.

Mathematical Finance and Asset Pricing Models | Feb 2023 – May 2023

Course project, Computational Finance: An Introduction to Option Pricing, IIT Goa

- Studied the Capital Asset Pricing Model (CAPM) and risk–return trade-offs, including the role of systematic risk in expected asset returns.
- Implemented Black–Scholes option pricing and derived analytical prices and Greeks under geometric Brownian motion.
- Developed Monte Carlo simulations for derivative pricing, validated them against closed-form solutions, and analysed option sensitivity to volatility, interest rates, and time to maturity for hedging.

EDUCATION

Bachelor of Technology in Mathematics and Computing | 2020 – 2024

Indian Institute of Technology (IIT) Goa, India

CGPA: 8.24 / 10 · Academic rank: 5th

Relevant coursework. Mathematics and finance: Linear Algebra, Probability and Statistics, Calculus, Differential Equations, Numerical Analysis, Computational Finance (Brownian motion, Itô's lemma, stochastic differential equations, Feynman–Kac, Black–Scholes), Complex Analysis, Discrete Mathematics. Computing and data: Algorithm Design, Data Structures and Algorithms, Econometric Data Science, Machine Learning, Theory of Computation, Introduction to Computing.

COURSES AND CERTIFICATIONS

Udemy and Coursera

- Introduction to Risk Management
- Introduction to Time Series and Forecasting
- Algorithmic Trading and Quantitative Analysis Using Python
- Mathematics for Machine Learning and Data Science Specialization

CONFERENCE PRESENTATION

Symposium on Data for Public Good, Indian Institute of Science (IISc) Bangalore | Nov 2023

Topic: Privacy-Preserving Synthetic Data using PATE-GAN

ACHIEVEMENTS

- Prime Minister's Scholarship Scheme (PMSS), Government of India, awarded for academic merit during undergraduate studies.
- Qualified JEE Advanced, the national entrance examination for the Indian Institutes of Technology.
- Captain, IIT Goa Football Team. Represented the institute at the Inter-IIT Sports Meet.
- State-level chess player. Secured 27th rank in the State Chess Championship.